

Day 1 — LLM & GenAI Interview Questions & Answers

44 interview-focused questions covering LLM fundamentals, tokens, context windows, parameters, Transformers, self-attention, training, RLHF, inference, and RAG.

LLM, Tokens, Context & Generation

1. What is an LLM?

Answer: An LLM, or Large Language Model, is a large neural-network model trained on massive amounts of text data to learn patterns in language and generate text.

2. What does “Large Language Model” mean?

Answer: Large refers to model size, training data, and computational scale. Language means it is designed to process and generate human language. Model means a mathematical/neural-network system that learns patterns from data.

3. Why is an LLM called “large”?

Answer: It is called large because it can contain billions of learned parameters and is trained using very large datasets and substantial computational resources.

4. What is a language model?

Answer: A language model learns patterns in language and estimates the probability of possible next tokens given the previous context.

5. What is a token?

Answer: A token is a unit of text processed by an LLM. It can be a complete word, part of a word, punctuation, or another piece of text.

6. Is a token the same as a word?

Answer: No. One word can correspond to one or multiple tokens, depending on the tokenizer and the language.

7. What is tokenization?

Answer: Tokenization is the process of breaking text into tokens that can be processed by the model.

8. What are input tokens?

Answer: Input tokens are the tokens contained in the information provided to the model, such as a user's prompt and other context.

9. What are output tokens?

Answer: Output tokens are the tokens generated by the model as its response.

10. What is a context window?

Answer: A context window is the maximum amount of token-based information that a model can consider in a single request.

11. What is the difference between parameters and tokens?

Answer: Parameters are learned numerical values inside the model, while tokens are units of text that the model processes or generates.

12. What is the difference between parameters and context window?

Answer: Parameters represent learned values inside the model, whereas the context window represents the maximum amount of token-based information the model can consider in one request.

13. What is the difference between tokens and context window?

Answer: A token is an individual unit of text, while the context window is the maximum number of tokens or amount of token-based information that can be considered in one request.

14. Why do tokens affect the cost of using an LLM?

Answer: Many LLM APIs charge based on the number of input and output tokens processed. Therefore, more tokens can result in higher usage costs.

15. Why do tokens affect context limits?

Answer: Because context windows are measured in tokens. As the number of tokens increases, more of the available context capacity is consumed.

16. What are parameters in an LLM?

Answer: Parameters are learned numerical values inside the neural network that are adjusted during training and collectively influence the model's behavior.

17. What does “a model has 7 billion parameters” mean?

Answer: It means the model contains approximately 7 billion learned numerical values. It does not mean the model contains exactly 7 billion individual facts.

18. What happens when you send a prompt to an LLM?

Answer: The prompt is tokenized, processed by the model/Transformer, a probability distribution over possible next tokens is produced, a token is selected, and this process repeats until a stopping condition is reached.

19. Does an LLM generate the entire answer at once?

Answer: No. An LLM generally generates the response autoregressively, predicting one token at a time based on the available context.

20. What is autoregressive generation?

Answer: Autoregressive generation means the model predicts the next token based on the tokens that came before it, including tokens it has already generated.

Transformers & Self-Attention

21. What is a Transformer?

Answer: A Transformer is a neural-network architecture that uses attention mechanisms to process relationships between tokens. It is the foundation of most modern LLMs.

22. Why were Transformers introduced?

Answer: Transformers addressed important limitations of recurrent models such as RNNs and LSTMs, particularly sequential processing and difficulty handling long-range relationships. They also enable efficient parallel processing during training and scale well.

23. What is attention?

Answer: Attention is a mechanism that allows a model to determine which parts of the input are more relevant when processing a particular token.

24. What is self-attention?

Answer: Self-attention allows each token to consider other tokens in the same sequence and determine how much information from them should influence its representation.

25. Why is self-attention powerful?

Answer: It allows the model to capture relationships between tokens, including tokens that may be far apart in a sequence, and helps the model understand context.

26. What is an encoder?

Answer: An encoder processes the input and creates contextual representations of it. It is primarily associated with understanding the input.

27. What is a decoder?

Answer: A decoder generates output, typically by predicting tokens based on the available context.

28. What is an encoder-decoder Transformer?

Answer: It contains both an encoder and decoder. The encoder processes the input, and the decoder uses that information to generate the output. A common use case is translation.

29. Are GPT-style models encoder-only, decoder-only, or encoder-decoder?

Answer: GPT-style models use a decoder-only Transformer architecture.

LLM Training & Inference

30. What is pre-training?

Answer: Pre-training is the initial large-scale training phase where an LLM learns general patterns of language from massive amounts of data, commonly through next-token prediction.

31. What is a base model?

Answer: A base model is the model obtained after pre-training before additional instruction tuning and alignment for assistant-style behavior.

32. What is instruction tuning?

Answer: Instruction tuning is additional training using instruction-response examples to teach a model to better follow user instructions.

33. What is RLHF?

Answer: RLHF stands for Reinforcement Learning from Human Feedback. It uses human preferences about model outputs to improve the model's behavior and alignment.

34. What is inference?

Answer: Inference is the process of using a trained model to generate an output for new input.

35. What is the difference between training and inference?

Answer: Training is when the model learns and its parameters are adjusted. Inference is when the trained model is used to generate predictions or outputs for new inputs.

Tricky / Conceptual Questions

36. Does an LLM store one answer for every question?

Answer: No. An LLM learns patterns in its parameters and generates responses based on the input context. It is not simply a database containing one answer for every possible question.

37. Does one parameter represent one fact?

Answer: No. Knowledge and behavior are distributed across many parameters. A parameter is simply a learned numerical value.

38. If a model has 70 billion parameters, does it have 70 billion pieces of knowledge?

Answer: No. Parameters are learned numerical values that collectively encode patterns and relationships. They should not be interpreted as individual facts.

39. If an LLM predicts “Delhi” with 95% probability, does that mean it knows the answer is definitely true?

Answer: No. The probability represents the model's prediction based on its learned patterns and current context. High probability does not guarantee factual correctness.

40. Why can an LLM hallucinate?

Answer: Because an LLM fundamentally generates text based on learned patterns and probabilities. It can produce a plausible-looking answer even when the information is incorrect or unsupported.

41. Why can an LLM generate a sentence it has never seen before?

Answer: Because it learns general patterns and relationships in language and can combine those learned patterns to generate new sequences of tokens.

42. Why can long conversations become expensive?

Answer: As conversation history grows, more tokens may need to be included and processed as context. More processed tokens can increase usage cost and consume more of the model's context window.

43. Why is RAG useful?

Answer: RAG retrieves relevant information from a large external knowledge base and provides that information to the LLM as context instead of sending the entire knowledge base to the model.

44. Is RAG the same as training the LLM?

Answer: No. Training learns or updates model parameters. RAG retrieves relevant external information and adds it to the context at inference time; it generally does not modify the model's parameters.

Quick Revision: Parameters = learned numerical values. Tokens = units of text processed/generated. Context window = maximum token-based information the model can consider in one request.